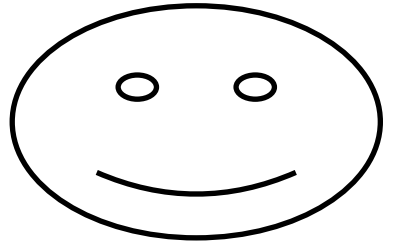


operators العمليات في الكمبيوتر

Operator types



Mathematical

$+$, $-$, $*$, $/$, $^$, mode

Relational

$<$, $>$, $<=$, $>=$,
 $==$ (is equal),
 $!=$ (not equal) ,
 $<>$ (not equal)

Logical

And , or , not

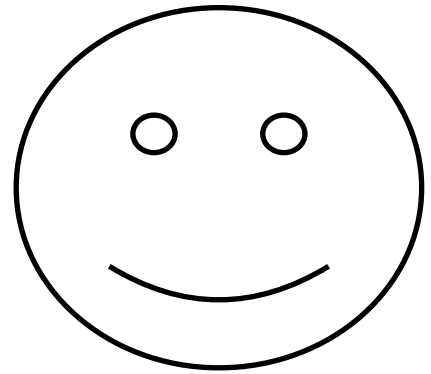
Mathematical operators —>

- 1) addition (+)
- 2) subtraction (-)
- 3) multiplication (*)
- 4) integer division (/)
- (إذا كان البسط او المقام عشري ف الناتج عشري)
- 5) modulo division (mod)
- 6) power (^)

6 + 5 معناه واحد بكل لغات البرمجة لكن يختلفوا ب السينتاكس الخاص فيهم من لغة الى اخرى

Relational operators —>

- 1) equal to (==)
- 2) not equal to (!= , < >)
- 3) less than (<)
- 4) grater than (>)
- 5) less than or equal (<=)
- 6) grater than or equal (>=)

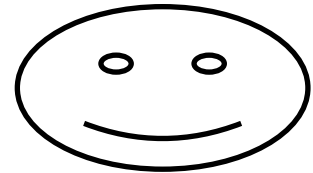


Logical operators

- 1) not (!)
- 2) and (&&)
- 3) or (||)

True = 1 (all number except 0)

False = 0 just



And operator

First condition	2nd condition	The result
True	True	True
True	False	False
False	True	False
False	False	False

Any “**false**” in “**and**” the “**result**” is **false**

حتى تكون النتيجة (true) لازم كل الشروط تتحقق ولا غير ذلك فالنتيجة هي (false)

Not operators

condition	! condition	The result
True	! True	False
False	! False	True

بتقوم بعكس النتيجة

Or operators

First condition	2nd condition	The result
True	True	True
True	False	True
False	True	True
False	False	False

Any “true” in “or” the result is “true”

في “or” بكفي ان يتحقق شرط واحد حتى تكون النتيجة “true”

Homework —>

Solve the following problems —>

1) Not (8 < 6) —> ! False —> true

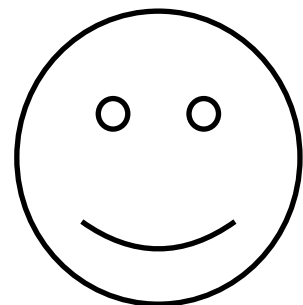
2) Not (8 == 8) —> ! True —> false

3) Not 0 —> ! False —> true

4) 12 >= 12 —> true

5) 8 < 6 —> false

6) 7 == 5 —> false



7) $8 == 8 \longrightarrow \text{true}$

8) $12 > 7 \longrightarrow \text{true}$

9) $0 \text{ or } 1 \longrightarrow \text{false} \parallel \text{true} \longrightarrow \text{true}$

10) $0 \text{ or } 0 \longrightarrow \text{false} \parallel \text{false} \longrightarrow \text{false}$

11) $1 \text{ and } 1 \longrightarrow \text{true} \&\& \text{true} \longrightarrow \text{true}$

12) $\text{Not } (7 == 5) \longrightarrow ! \text{ False} \longrightarrow \text{true}$

13) $12 \leq 12 \longrightarrow \text{true}$

14) $\text{Not } (1 \text{ or } 0) \longrightarrow ! (\text{true} \parallel \text{false}) \longrightarrow ! \text{ True} \longrightarrow \text{false}$

15) $\text{Not } (12 \leq 12) \longrightarrow ! \text{ True} \longrightarrow \text{false}$

16) $\text{true and } 0 \longrightarrow \text{true} \&\& \text{false} \longrightarrow \text{false}$

17) $\text{Not } (12 \geq 12) \longrightarrow ! \text{ True} \longrightarrow \text{false}$

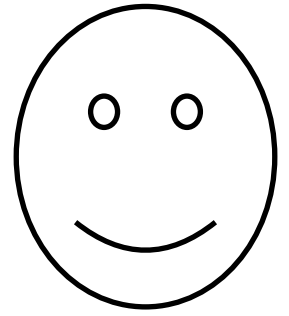
18) $(7 == 7) \text{ and } (7 > 5) \longrightarrow \text{true} \&\& \text{true} \longrightarrow \text{true}$

19) $(7 == 7) \text{ and } (7 < 5) \longrightarrow \text{true} \&\& \text{false} \longrightarrow \text{false}$

20) $(7 == 7) \text{ or } (7 < 5) \longrightarrow \text{true} \parallel \text{false} \longrightarrow \text{true}$

21) $\text{Not } (12 < 7) \longrightarrow ! \text{ False} \longrightarrow \text{true}$

22) $\text{Not } (8 < 6) \longrightarrow ! \text{ False} \longrightarrow \text{true}$



23) $(7 < 7) \text{ or } (7 > 5) \longrightarrow \text{false} \parallel \text{true} \longrightarrow \text{true}$

24) $\text{Not } (7 == 7) \text{ and } (7 > 5) \longrightarrow !\text{True} \ \&\& \ \text{true} \longrightarrow \text{False} \ \&\& \ \text{true} \longrightarrow \text{false}$

25) $(7 == 7) \text{ and } \text{Not}(7 < 5) \longrightarrow \text{true} \ \&\& \ !\text{false} \longrightarrow \text{true} \ \&\& \ \text{true} \longrightarrow \text{true}$

Calculation priorities اولويات الحساب

الاقواس ب التعامل معهم بكون من
الداخل الى الخارج

Order of Operations

B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract
()	$\sqrt{x} \ x^2$	$\div$ or $\times$	$+$ or $-$		
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S



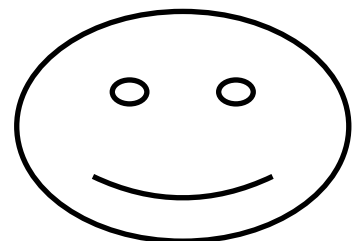
From Left to Right

$4 * 4 + 4 * 4 + 4 - 4 * 4 \longrightarrow$

$(4*4)+(4*4) + 4 - (4*4) = (16 + 16)+ 4 - 16 =$

$(32 + 4) - 16 = 36 - 16 =$

20



$$\cdot 18 / (9 - 2 * 3) =$$

$$18 / (9 - 6) =$$

$$18 / 3 = 6$$

$$\cdot 10 - [72 / (4 + 5)] =$$

$$10 - [72 / 9] =$$

$$10 - 8 = 2$$

$$\cdot (6 * 2 - 6 - 1) * 2^2 =$$

$$(12 - 6 - 1) * 2^2 =$$

$$(6 - 1) * 2^2 =$$

$$5 * 2^2 =$$

$$5 * 4 = 20$$

$$\cdot 4 * (2 + (7 * (5 - 3))) =$$

$$4 * (2 + (7 * 2)) =$$

$$4 * (2 + 14) =$$

$$4 * 16 = 64$$

$$(1 + 20 - 9 / 3^2) / ((3 + 1)^2 + 8/2) =$$

$$(1 + 20 - 9 / 9) / ((3 + 1)^2 + 8/2) =$$

$$(1 + 20 - 1) / ((3 + 1)^2 + 8/2) =$$

$$(21 - 1) / ((3 + 1)^2 + 8/2) =$$

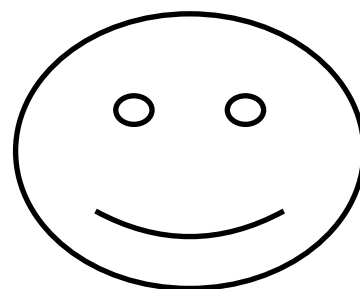
$$20 / ((3 + 1)^2 + 8/2) =$$

$$20 / (4^2 + 8/2) =$$

$$20 / (16 + 8/2) =$$

$$20 / (16 + 4) =$$

$$20 / 20 = 1$$



Logic priorities —>

True and ((9>2) or (4+2 == 5))=

True and (true or (4+2 == 5)) =

True and (true or (6 == 5)) =

True and (true or false) =

True and true = true

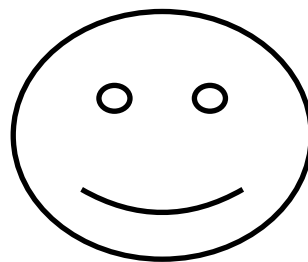
Home work —>

Solve the following problems —>

1) $6 / 3 + 4 * 2 =$

$2 + 4 * 2 =$

$2 + 8 = 10$



2) $6 * 4 - 12 / 3 - 8 =$

$24 - 12 / 3 - 8 =$

$24 - 4 - 8 =$

$20 - 8 = 12$

3) $4 * 4 - 3 * 3 - 16 / 4 =$

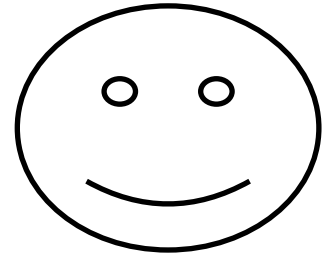
$16 - 3 * 3 - 16 / 4 =$

$16 - 9 - 16 / 4 =$

$16 - 9 - 4 =$

$7 - 4 = 3$

$$\begin{aligned}
 4) \quad & 20 - (3 * 23 - 5) = \\
 & 20 - (69 - 5) = \\
 & 20 - 64 = -44
 \end{aligned}$$

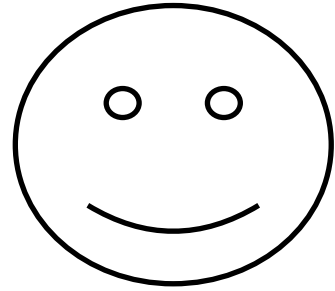


$$\begin{aligned}
 5) \quad & (5+2)^2 - 9 * 3 + 2^3 = \\
 & 7^2 - 9 * 3 + 2^3 = \\
 & 49 - 9 * 3 + 2^3 = \\
 & 49 - 9 * 3 + 8 = \\
 & 49 - 27 + 8 = \\
 & 22 + 8 = 30
 \end{aligned}$$

$$\begin{aligned}
 6) \quad & 5 * (2^2 + 3) + (2^3)^2 = \\
 & 5 * (4 + 3) + (2^3)^2 = \\
 & 5 * 7 + (2^3)^2 = \\
 & 5 * 7 + 8^2 = \\
 & 5 * 7 + 64 = \\
 & 35 + 64 = 99
 \end{aligned}$$

$$\begin{aligned}
 7) \quad & (5 > 6 \text{ and } 7 == 7) \text{ or } (1 \text{ or } 0) = \\
 & (5 > 6 \ \&\& \ 7 == 7) \ || \ (1 \ || \ 0) = \\
 & (\text{false} \ \&\& \ 7 == 7) \ || \ (1 \ || \ 0) = \\
 & (\text{false} \ \&\& \ \text{true}) \ || \ (1 \ || \ 0) = \\
 & \text{false} \ || \ (1 \ || \ 0) = \\
 & \text{false} \ || \ (\text{true} \ || \ \text{false}) = \\
 & \text{false} \ || \ \text{true} = \text{true}
 \end{aligned}$$

$$\begin{aligned}
 8) \quad & (7 - \sqrt{9}) * (42 - 3 + 1) = \\
 & (7 - 3) * (42 - 3 + 1) = \\
 & 4 * (42 - 3 + 1) = \\
 & 4 * (39 + 1) = \\
 & 4 * 40 = 160
 \end{aligned}$$



$$\begin{aligned}
 9) \quad & 6 * 2 + 5 * 1 + 4 / 2 - 1 = \\
 & 12 + 5 * 1 + 4 / 2 - 1 = \\
 & 12 + 5 + 4 / 2 - 1 = \\
 & 12 + 5 + 2 - 1 = \\
 & 17 + 2 - 1 = \\
 & 19 - 1 = 18
 \end{aligned}$$

$$\begin{aligned}
 10) \quad & \text{not } (5 > 6 \text{ and } 7 == 7) \text{ or } (1 \text{ or } 0) = \\
 & ! (5 > 6 \ \&\& \ 7 == 7) \parallel (\text{true} \parallel \text{false}) = \\
 & ! (\text{false} \ \&\& \ \text{true}) \parallel (\text{true} \parallel \text{false}) = \\
 & ! \text{False} \parallel (\text{true} \parallel \text{false}) = \\
 & ! \text{false} \parallel \text{true} = \\
 & \text{True} \parallel \text{true} = \text{true}
 \end{aligned}$$

$$\begin{aligned}
 11) \quad & (2 + 1 - 3 + 1 + 4) ^2 - 2 * 4 = \\
 & (3 - 3 + 1 + 4) ^2 - 2 * 4 = \\
 & (0 + 1 + 4) ^2 - 2 * 4 = \\
 & (1 + 4) ^2 - 2 * 4 = \\
 & 5^2 - 2 * 4 = \\
 & 25 - 2 * 4 = \\
 & 25 - 8 = 17
 \end{aligned}$$

12) $((5 > 6 \text{ and not}(7 \leq 8)) \text{ and } (8 > 1 \text{ or } 4 \leq 3)) \text{ or true}$

$((5 > 6 \ \&\& \ ! (7 \leq 8)) \ \&\& \ (8 > 1 \ \parallel \ 4 \leq 3)) \parallel \text{true} =$
 $((5 > 6 \ \&\& \ ! (\text{true})) \ \&\& \ (8 > 1 \ \parallel \ 4 \leq 3)) \parallel \text{true} =$
 $((5 > 6 \ \&\& \ \text{false}) \ \&\& \ (8 > 1 \ \parallel \ 4 \leq 3)) \parallel \text{true} =$
 $((\text{false} \ \&\& \ \text{false}) \ \&\& \ (8 > 1 \ \parallel \ 4 \leq 3)) \parallel \text{true} =$
 $(\text{false} \ \&\& \ (8 > 1 \ \parallel \ 4 \leq 3)) \parallel \text{true} =$
 $(\text{false} \ \&\& \ (\text{true} \ \parallel \ 4 \leq 3)) \parallel \text{true} =$
 $(\text{false} \ \&\& \ (\text{true} \ \parallel \ \text{false})) \parallel \text{true} =$
 $(\text{false} \ \&\& \ \text{true}) \parallel \text{true} =$
 $\text{false} \parallel \text{true} = \text{true}$

13) $1 + (2 * (4 - 3 + 1) + 7) - 2 =$

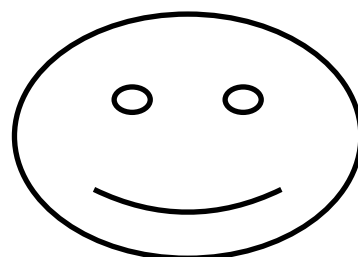
$1 + (2 * (1 + 1) + 7) - 2 =$

$1 + (2 * 2 + 7) - 2 =$

$1 + (4 + 7) - 2 =$

$1 + 11 - 2 =$

$12 - 2 = 10$



14) $(20 - 18)^3 / 8 * 3 - 1 =$

$2^3 / 8 * 3 - 1 =$

$8 / 8 * 3 - 1 =$

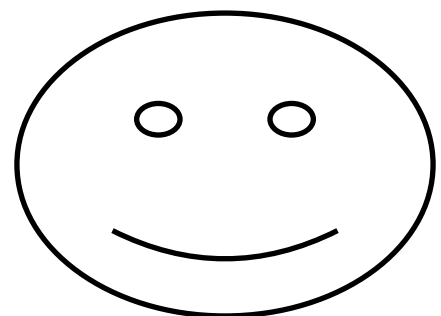
$1 * 3 - 1 =$

$3 - 1 = 2$

15) not (5 > 6 and 7==7) or not (true||false)=
 ! (5 > 6 && 7 == 7) || ! (true || false)=
 ! (false && 7 == 7) || ! (true || false)=
 ! (false && true) || ! (true || false)=
 ! false || ! (true || false) =
 ! false || ! true =
 true || ! true =
 true || false = true

16) ((5 > 6 and 7<= 8) or (8>1 and 4<=3)) and
 true
 ((5>6&&7<= 8) || (8>1 && 4<=3)) && true
 ((false&& 7<= 8) ||(8>1 && 4<=3)) &&true
 ((false && true) ||(8>1 && 4<=3)) &&true
 (false || (8>1 && 4<=3)) && true
 (false || (true && 4<=3)) && true
 (false || (true && false)) && true
 (false || false) && true
 false && true = false

17) 10 + 6 * (3-2) * 8 / 24 - 1 =
 10 + 6* 1 * 8 / 24 - 1 =
 10 + 6 * 8 / 24 - 1 =
 10 + 48 / 24 - 1 =
 10 + 2 - 1 =
 12 - 1 = 11



18) Not (5 > 6 or 7 == 7) and not (1 or 0) =
!(5>6 || 7 ==7) && !(1or 0) =
!(5>6 || 7 ==7) && !(true or false) =
!(false || true) && !(true or false) =
! true && !(true or false) =
! true && ! true =
False && false = false

